

MATPREDICT · MATING-TYPE REPORT

basidio_microbotryum_nocall

taxid 683840 · Basidiomycota

Run by MATPredict 0.6.1 · database 2b90ce832408 · NCBI E-utilities · rendered 2026-10-08 04:26 UTC by MATPredict 0.6.1

Result

No MAT locus called

No family reached the evidence bar. That is weak evidence that the locus is absent: a fragmented assembly, or a lineage far from every reference, often explains it.

Next: check the withheld candidate loci below against a known locus, or type the strain from its reads.

What was searched

ROUTING

Whole phylum. No curated family covers this lineage, so every family of the genome's phylum was searched. Family labels are less reliable here.

FAMILIES SEARCHED	GENETIC CODE	TAXONOMY
Basidiomycota:HD, Basidiomycota:PR, Basidiomycota:MAT, Basidiomycota:Aalpha, Basidiomycota:Abeta, Basidiomycota:Balpha, Basidiomycota:Bbeta, Basidiomycota:aLocus, Basidiomycota:bLocus	1	NCBI E-utilities

FAMILIES SEARCHED BUT NOT CALLED

family	reason	best % of genes found	genes found
Basidiomycota:HD	best cluster carried 0 modelled gene(s), below the 2 required to report a locus; none of its genes could be modelled	100.0	HD2
Basidiomycota:PR	best cluster carried 0 modelled gene(s), below the 2 required to report a locus; none of its genes could be modelled	50.0	pheromone_B44
Basidiomycota:MAT	best cluster carried 0 modelled gene(s), below the 2 required to report a locus; none of its genes could be modelled	33.3	SXI2
Basidiomycota:Aalpha	best cluster carried 0 modelled gene(s), below the 2 required to report a locus; none of its genes could be modelled	50.0	Y

family	reason	best % of genes found	genes found
Basidiomycota:Abeta	no reference protein for this family is in the search set (none curated, or all withheld)	0.0	-
Basidiomycota:Balpha	best cluster carried 0 modelled gene(s), below the 2 required to report a locus; none of its genes could be modelled	50.0	bar3
Basidiomycota:Bbeta	best cluster carried 0 modelled gene(s), below the 2 required to report a locus; none of its genes could be modelled	25.0	bbp2_a
Basidiomycota:aLocus	best cluster carried 0 modelled gene(s), below the 2 required to report a locus; none of its genes could be modelled	66.7	pra1
Basidiomycota:bLocus	best cluster carried 1 modelled gene(s), below the 2 required to report a locus; only 1 of its genes could be modelled	100.0	bE, bW

PHEROMONE-RECEPTOR ARRAYS

Receptor arrays occur for both mating and non-mating STE3-like receptors (the study found arrays holding paralogs beside the mating receptors in *Coprinopsis cinerea* and *Schizophyllum commune*), so array membership does not establish that a locus is a mating receptor. `receptor_array_support` is a flag on the evidence behind an array; it never changes a call, its confidence or its verification label.

array	location	loci	calls	support	reasons
Basidiomycota:PR:GL541857.1:1877-3126	GL541857.1:1,877-3,126	1	0	unsupported	a single receptor locus; no pheromone-precursor homology hit; 0 strict-CAAX pheromone ORFs

Withheld candidate loci 303

Places with some MAT-gene evidence that did not pass a quality gate. Most genomes have several. They are listed so a known locus can be checked against them, not because they are likely calls.

Reasons: fewer than 2 genes could be modelled (303).

WITHHELD LOCI (303)

family	location	idiomorph	genes	modelled	reason
HD	GL541668.1:765-22,935	undetermined	HD2	0	fewer than 2 genes could be modelled
Aalpha	GL541668.1:765-22,935	undetermined	Y	0	fewer than 2 genes could be modelled
bLocus	GL541668.1:765-22,935	undetermined	bW	0	fewer than 2 genes could be modelled
MAT	GL541668.1:765-22,935	a	SXI2	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
HD	GL541668.1: 181,210–181,389	undetermined	HD2	0	fewer than 2 genes could be modelled
aLocus	GL541668.1: 216,319–216,408	undetermined	pra1	0	fewer than 2 genes could be modelled
HD	GL541860.1: 3,368–5,616	undetermined	HD1, HD2	0	fewer than 2 genes could be modelled
bLocus	GL541860.1: 3,368–5,616	undetermined	bE, bW	1	fewer than 2 genes could be modelled
Aalpha	GL541860.1: 3,368–5,616	undetermined	Z	0	fewer than 2 genes could be modelled
MAT	GL541860.1: 3,368–5,616	alpha	SXI1	0	fewer than 2 genes could be modelled
HD	GL541684.1: 150,669–150,893	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL541735.1: 18,750–60,889	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541735.1: 18,750–60,889	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541735.1: 18,750–60,889	undetermined	mfa1	0	fewer than 2 genes could be modelled
Bbeta	GL541735.1: 18,750–60,889	undetermined	bbp2_a	0	fewer than 2 genes could be modelled
bLocus	GL541687.1: 46,380–46,523	undetermined	bE	0	fewer than 2 genes could be modelled
HD	GL541687.1: 73,034–86,590	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541687.1: 168,082–168,195	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541659.1: 89,715–107,940	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541659.1: 179,713–179,922	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541659.1: 210,996–258,054	undetermined	HD2	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541659.1: 210,996–258,054	undetermined	bW	0	fewer than 2 genes could be modelled
PR	GL541659.1: 210,996–258,054	undetermined	pheromone_B44	0	fewer than 2 genes could be modelled
Bbeta	GL541659.1: 210,996–258,054	undetermined	bbp2-8, bbp2_b	0	fewer than 2 genes could be modelled
HD	GL541782.1: 15,213–15,341	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL541663.1: 92,235–103,773	undetermined	HD1	0	fewer than 2 genes could be modelled
aLocus	GL541663.1: 136,673–161,137	undetermined	mfa1, rba1	0	fewer than 2 genes could be modelled
bLocus	GL541646.1: 187,432–187,758	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541646.1: 285,028–285,171	undetermined	bE	0	fewer than 2 genes could be modelled
Aalpha	GL541646.1: 339,548–348,289	undetermined	Y	0	fewer than 2 genes could be modelled
bLocus	GL541646.1: 339,548–348,289	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541646.1: 461,139–461,303	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541646.1: 545,845–554,175	undetermined	bE, bW	0	fewer than 2 genes could be modelled
HD	GL541646.1: 545,845–554,175	undetermined	HD1	0	fewer than 2 genes could be modelled
MAT	GL541646.1: 545,845–554,175	alpha	SXI1	0	fewer than 2 genes could be modelled
bLocus	GL541664.1: 11,881–12,066	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541664.1: 44,063–48,907	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541664.1: 165,344–170,760	undetermined	HD1	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
aLocus	GL541672.1: 11,220–20,868	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541672.1: 89,200–95,249	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541672.1: 176,643–176,810	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541672.1: 205,466–213,463	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541691.1: 68,885–125,372	undetermined	bE, bW	0	fewer than 2 genes could be modelled
aLocus	GL541691.1: 68,885–125,372	undetermined	pra1	0	fewer than 2 genes could be modelled
MAT	GL541691.1: 68,885–125,372	a	SXI2	0	fewer than 2 genes could be modelled
HD	GL541691.1: 68,885–125,372	undetermined	MIP1	0	fewer than 2 genes could be modelled
HD	GL541675.1: 49,104–77,417	undetermined	HD2	0	fewer than 2 genes could be modelled
Aalpha	GL541675.1: 49,104–77,417	undetermined	Z	0	fewer than 2 genes could be modelled
PR	GL541675.1: 49,104–77,417	undetermined	pheromone_receptor	0	fewer than 2 genes could be modelled
HD	GL541675.1: 111,786–131,229	undetermined	HD1, MIP1	0	fewer than 2 genes could be modelled
bLocus	GL541675.1: 162,361–176,416	undetermined	bE	0	fewer than 2 genes could be modelled
Balpha	GL541857.1: 1,805–13,227	undetermined	bar3	0	fewer than 2 genes could be modelled
aLocus	GL541857.1: 1,805–13,227	undetermined	pra1	0	fewer than 2 genes could be modelled
MAT	GL541857.1: 1,805–13,227	undetermined	STE3	0	fewer than 2 genes could be modelled
PR	GL541857.1: 1,805–13,227	undetermined	pheromone_receptor	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
Bbeta	GL541857.1: 1,805–13,227	undetermined	bbr2	0	fewer than 2 genes could be modelled
HD	GL541857.1: 1,805–13,227	undetermined	MIP1	0	fewer than 2 genes could be modelled
bLocus	GL541709.1: 60,444–60,824	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541725.1: 22,542–44,103	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	GL541725.1: 99,383–99,607	undetermined	bE	0	fewer than 2 genes could be modelled
Aalpha	GL541704.1: 68,404–68,631	undetermined	Z	0	fewer than 2 genes could be modelled
HD	GL541797.1: 12,736–13,023	undetermined	HD1, HD2	0	fewer than 2 genes could be modelled
Aalpha	GL541797.1: 12,736–13,023	undetermined	Y	0	fewer than 2 genes could be modelled
bLocus	GL541643.1: 87,141–93,907	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541643.1: 281,650–299,701	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541643.1: 510,871–511,125	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541643.1: 582,216–582,440	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541643.1: 654,678–654,932	undetermined	bE	0	fewer than 2 genes could be modelled
HD	GL541643.1: 690,514–690,735	undetermined	HD1	0	fewer than 2 genes could be modelled
Aalpha	GL541643.1: 742,167–742,379	undetermined	Y	0	fewer than 2 genes could be modelled
Balpha	GL541643.1: 777,664–777,786	undetermined	bap3-3	0	fewer than 2 genes could be modelled
bLocus	GL541647.1: 58,664–58,948	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
HD	GL541647.1: 89,363–113,024	undetermined	HD1, MIP1	0	fewer than 2 genes could be modelled
bLocus	GL541647.1: 89,363–113,024	undetermined	bW	0	fewer than 2 genes could be modelled
PR	GL541647.1: 89,363–113,024	undetermined	fungal_mating_type_pheromone	0	fewer than 2 genes could be modelled
Balpha	GL541647.1: 143,203–143,301	undetermined	bap3-3	0	fewer than 2 genes could be modelled
bLocus	GL541647.1: 188,191–190,065	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	GL541647.1: 223,101–223,259	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541647.1: 354,297–366,785	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541647.1: 425,762–425,944	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541670.1: 446–21,962	undetermined	bE, bW	0	fewer than 2 genes could be modelled
HD	GL541670.1: 446–21,962	undetermined	HD2	0	fewer than 2 genes could be modelled
aLocus	GL541670.1: 446–21,962	undetermined	rba1	0	fewer than 2 genes could be modelled
aLocus	GL541670.1: 119,902–137,933	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541670.1: 119,902–137,933	undetermined	bW	0	fewer than 2 genes could be modelled
Balpha	GL541670.1: 190,463–190,591	undetermined	bap3-3	0	fewer than 2 genes could be modelled
Balpha	GL541665.1: 166,052–177,166	undetermined	bap3-3	0	fewer than 2 genes could be modelled
Balpha	GL541722.1: 52,349–52,459	undetermined	bap3-3	0	fewer than 2 genes could be modelled
Balpha	GL541830.1: 10,834–13,806	undetermined	bap3-3	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541830.1: 10,834–13,806	undetermined	bE	0	fewer than 2 genes could be modelled
Balpha	GL541694.1: 94,597–94,764	undetermined	bap3-3	0	fewer than 2 genes could be modelled
HD	GL541694.1: 143,415–143,552	undetermined	HD1	0	fewer than 2 genes could be modelled
Balpha	GL541645.1: 91,341–91,436	undetermined	bap3-3	0	fewer than 2 genes could be modelled
bLocus	GL541645.1: 307,584–326,915	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541645.1: 396,897–416,087	undetermined	HD2	0	fewer than 2 genes could be modelled
PR	GL541645.1: 396,897–416,087	undetermined	pheromone_receptor, fungal_mating_type_pheromone	0	fewer than 2 genes could be modelled
aLocus	GL541645.1: 458,242–458,370	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541645.1: 539,325–543,911	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541645.1: 581,258–595,946	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	GL541645.1: 581,258–595,946	undetermined	bW	0	fewer than 2 genes could be modelled
Bbeta	GL541645.1: 581,258–595,946	undetermined	bbp2_a	0	fewer than 2 genes could be modelled
bLocus	GL541789.1: 18,217–28,119	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541654.1: 179,554–198,786	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541654.1: 257,559–267,754	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541649.1: 43,238–43,399	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541702.1: 15,226–42,666	undetermined	bE, bW	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541702.1: 104,794–104,925	undetermined	bE	0	fewer than 2 genes could be modelled
Balpha	GL541858.1: 7,677–7,907	undetermined	bar3	0	fewer than 2 genes could be modelled
HD	GL541705.1: 15,593–21,268	undetermined	HD1	0	fewer than 2 genes could be modelled
Balpha	GL541705.1: 15,593–21,268	undetermined	bar3	0	fewer than 2 genes could be modelled
bLocus	GL541705.1: 124,233–124,433	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541685.1: 85,006–104,705	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541682.1: 51,392–72,048	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541732.1: 40,794–70,775	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541700.1: 23,933–24,076	undetermined	bE	0	fewer than 2 genes could be modelled
PR	GL541700.1: 86,506–110,209	undetermined	pheromone_receptor, pheromone_B44	0	fewer than 2 genes could be modelled
bLocus	GL541669.1: 10,134–25,742	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541669.1: 134,658–134,912	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541689.1: 28,152–32,926	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	GL541689.1: 90,758–114,038	undetermined	bE, bW	0	fewer than 2 genes could be modelled
PR	GL541689.1: 90,758–114,038	undetermined	pheromone_receptor, fungal_mating_type_pheromone	0	fewer than 2 genes could be modelled
Bbeta	GL541689.1: 90,758–114,038	undetermined	bbp2-1	0	fewer than 2 genes could be modelled
aLocus	GL541689.1: 145,530–159,223	undetermined	pra1	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
HD	GL541655.1: 191,957–195,055	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541764.1: 38,807–38,935	undetermined	bE	0	fewer than 2 genes could be modelled
HD	GL541674.1: 6,875–13,341	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541652.1: 214,343–214,588	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	GL541746.1: 30,179–57,468	undetermined	mfa1, pra1	0	fewer than 2 genes could be modelled
bLocus	GL541746.1: 30,179–57,468	undetermined	bW	0	fewer than 2 genes could be modelled
PR	GL541746.1: 30,179–57,468	undetermined	pheromone_receptor	0	fewer than 2 genes could be modelled
Bbeta	GL541746.1: 30,179–57,468	undetermined	bbp2_a	0	fewer than 2 genes could be modelled
bLocus	GL541733.1: 13,902–30,168	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541695.1: 59,205–67,087	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541657.1: 10,635–10,880	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541657.1: 179,592–223,042	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541657.1: 179,592–223,042	undetermined	bW	0	fewer than 2 genes could be modelled
PR	GL541657.1: 179,592–223,042	undetermined	pheromone_receptor	0	fewer than 2 genes could be modelled
Bbeta	GL541657.1: 179,592–223,042	undetermined	bbp2_b	0	fewer than 2 genes could be modelled
aLocus	GL541651.1: 32,320–32,466	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541651.1: 184,522–191,489	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541651.1: 224,378–224,476	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541651.1: 326,920–327,048	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541749.1: 8,808–34,304	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541688.1: 7,285–7,398	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541688.1: 95,794–113,087	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL541719.1: 42,760–53,790	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541660.1: 11,634–11,855	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541660.1: 91,133–97,909	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	GL541660.1: 141,938–142,941	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541741.1: 55,615–55,716	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541644.1: 9,243–61,326	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541644.1: 9,243–61,326	undetermined	bE	0	fewer than 2 genes could be modelled
PR	GL541644.1: 9,243–61,326	undetermined	pheromone_receptor	0	fewer than 2 genes could be modelled
HD	GL541644.1: 127,631–173,661	undetermined	HD1	0	fewer than 2 genes could be modelled
aLocus	GL541644.1: 127,631–173,661	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541644.1: 127,631–173,661	undetermined	bW	0	fewer than 2 genes could be modelled
PR	GL541644.1: 127,631–173,661	undetermined	pheromone_receptor	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541644.1: 231,569–231,793	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541644.1: 408,254–408,424	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541644.1: 496,692–501,978	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541644.1: 535,059–541,013	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541644.1: 626,328–646,413	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541698.1: 2,244–2,438	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541698.1: 110,091–130,606	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541727.1: 43,330–43,644	undetermined	pra1	0	fewer than 2 genes could be modelled
HD	GL541727.1: 77,444–96,525	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541727.1: 77,444–96,525	undetermined	bE	0	fewer than 2 genes could be modelled
MAT	GL541727.1: 77,444–96,525	undetermined	STE3	0	fewer than 2 genes could be modelled
Bbeta	GL541727.1: 77,444–96,525	undetermined	bbp2-6	0	fewer than 2 genes could be modelled
HD	GL541717.1: 44,235–50,674	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541717.1: 88,716–99,107	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541658.1: 22,668–22,793	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL541681.1: 42,174–51,121	undetermined	HD2	0	fewer than 2 genes could be modelled
aLocus	GL541681.1: 42,174–51,121	undetermined	pra1	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541681.1: 92,754–93,017	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541681.1: 169,312–191,840	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541667.1: 70,365–74,831	undetermined	bE, bW	0	fewer than 2 genes could be modelled
HD	GL541667.1: 161,504–161,641	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541667.1: 161,504–161,641	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	GL541667.1: 232,763–233,292	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541661.1: 3,163–39,617	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541661.1: 142,768–142,908	undetermined	pra1	0	fewer than 2 genes could be modelled
HD	GL541661.1: 243,011–243,142	undetermined	HD2	0	fewer than 2 genes could be modelled
HD	GL541766.1: 6,498–24,176	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541653.1: 147,887–148,042	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541653.1: 240,173–240,418	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541653.1: 331,179–331,370	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541726.1: 48,595–62,401	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541683.1: 23,397–40,423	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	GL541683.1: 86,848–102,755	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541683.1: 170,026–170,148	undetermined	HD1	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
PR	GL541738.1: 7,687–18,821	undetermined	pheromone_B43, fungal_mating_type_pheromone	0	fewer than 2 genes could be modelled
bLocus	GL541738.1: 7,687–18,821	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541739.1: 59,701–59,790	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	GL541648.1: 143,736–143,924	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541648.1: 252,449–252,547	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541648.1: 353,617–353,745	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541648.1: 394,446–412,355	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541699.1: 137,843–138,022	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541656.1: 3,705–23,493	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	GL541656.1: 3,705–23,493	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541656.1: 185,165–185,278	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541656.1: 232,888–233,022	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541656.1: 319,297–319,697	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	GL541807.1: 5,554–14,483	undetermined	pra1	0	fewer than 2 genes could be modelled
aLocus	GL541662.1: 14,332–32,232	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	GL541662.1: 59,367–59,750	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541662.1: 224,670–224,795	undetermined	bW	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
HD	GL541936.1: 925–1,032	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL541673.1: 91,102–106,002	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL541849.1: 5,362–6,492	undetermined	HD2	0	fewer than 2 genes could be modelled
aLocus	GL541849.1: 5,362–6,492	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541745.1: 20,564–20,695	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541650.1: 23,268–23,432	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	GL541650.1: 87,438–87,569	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541650.1: 123,894–124,091	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541650.1: 216,514–236,883	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	GL541650.1: 305,200–305,409	undetermined	bW	0	fewer than 2 genes could be modelled
PR	GL541650.1: 371,539–387,034	undetermined	pheromone_receptor, fungal_mating_type_pheromone	0	fewer than 2 genes could be modelled
bLocus	GL541650.1: 371,539–387,034	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541761.1: 5,639–5,758	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541761.1: 49,095–49,741	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541829.1: 10,983–11,102	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541790.1: 8,267–8,401	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541865.1: 12,405–12,530	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541808.1: 32,480–32,644	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541916.1: 3,539–3,811	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541676.1: 91,616–91,750	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541676.1: 161,000–177,548	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541823.1: 4,260–11,288	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541786.1: 811–5,983	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541759.1: 17,756–40,238	undetermined	mfa1, pra1	0	fewer than 2 genes could be modelled
HD	GL541692.1: 31,597–31,773	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541692.1: 140,881–141,021	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541818.1: 13,037–13,198	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL541753.1: 60,251–60,412	undetermined	HD1	0	fewer than 2 genes could be modelled
bLocus	GL541678.1: 6,516–6,671	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541678.1: 93,049–111,239	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541810.1: 14,921–28,922	undetermined	pra1	0	fewer than 2 genes could be modelled
aLocus	GL541729.1: 5,971–37,020	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541729.1: 5,971–37,020	undetermined	bE	0	fewer than 2 genes could be modelled
MAT	GL541729.1: 5,971–37,020	undetermined	FAO1	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
aLocus	GL541752.1: 58,983–59,090	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	GL541757.1:238–738	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541851.1: 4,586–14,603	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541837.1: 4,190–13,193	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541697.1: 5,148–5,450	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	GL541697.1: 65,251–81,442	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541671.1: 19,644–19,733	undetermined	bW	0	fewer than 2 genes could be modelled
HD	GL541902.1: 3,610–3,825	undetermined	HD1	0	fewer than 2 genes could be modelled
HD	GL541802.1: 34,897–35,034	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541693.1: 85,368–85,457	undetermined	bE	0	fewer than 2 genes could be modelled
HD	GL541693.1: 122,899–138,650	undetermined	HD2	0	fewer than 2 genes could be modelled
bLocus	GL541693.1: 122,899–138,650	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541666.1: 22,882–23,031	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541666.1: 86,678–86,980	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541666.1: 142,487–142,675	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541706.1: 6,407–6,661	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541706.1: 80,084–80,188	undetermined	bW	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541787.1: 28,164–39,537	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	GL541826.1: 15,045–15,245	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541945.1: 2,131–2,223	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541712.1: 38,032–46,829	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	GL541827.1: 17,900–18,043	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541788.1: 29,084–29,200	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541777.1: 31,987–37,883	undetermined	bE, bW	0	fewer than 2 genes could be modelled
aLocus	GL541747.1: 63,413–63,556	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541716.1: 47,075–47,233	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541924.1:4–180	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541834.1: 5,701–5,877	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541926.1:427–603	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541714.1: 23,672–23,953	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541769.1: 40,546–40,758	undetermined	bW	0	fewer than 2 genes could be modelled
aLocus	GL541711.1: 2,946–3,095	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541711.1: 44,233–78,773	undetermined	bE, bW	1	fewer than 2 genes could be modelled
bLocus	GL541744.1: 22,879–23,010	undetermined	bE	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541744.1: 59,479–59,667	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541680.1: 41,801–41,956	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541680.1: 76,330–76,674	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541792.1: 24,572–24,676	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541886.1: 2,366–2,851	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541779.1: 13,256–13,411	undetermined	pra1	0	fewer than 2 genes could be modelled
bLocus	GL541686.1: 23,334–36,920	undetermined	bE, bW	0	fewer than 2 genes could be modelled
bLocus	GL541686.1: 108,543–108,641	undetermined	bE	0	fewer than 2 genes could be modelled
aLocus	GL541861.1: 10,555–10,740	undetermined	rba1	0	fewer than 2 genes could be modelled
aLocus	GL541832.1: 20,044–20,148	undetermined	rba1	0	fewer than 2 genes could be modelled
bLocus	GL541742.1: 18,930–19,193	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541820.1: 1,497–1,595	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541838.1: 4,562–4,699	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541794.1: 5,109–5,276	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541758.1: 12,829–12,984	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541754.1: 7,712–7,822	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541859.1: 14,118–14,300	undetermined	bW	0	fewer than 2 genes could be modelled

family	location	idiomorph	genes	modelled	reason
bLocus	GL541778.1: 18,784–18,909	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541703.1: 60,607–60,828	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541703.1: 100,901–101,011	undetermined	bW	0	fewer than 2 genes could be modelled
bLocus	GL541824.1: 1,706–13,535	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541765.1: 21,636–21,752	undetermined	bE	0	fewer than 2 genes could be modelled
bLocus	GL541809.1: 28,816–28,986	undetermined	bW	0	fewer than 2 genes could be modelled
Bbeta	GL541659.1: 210,996–258,054	undetermined	bbp2-8, bbp2_b	0	fewer than 2 genes could be modelled
Bbeta	GL541679.1: 112,869–163,230	undetermined	bbp2-2, bbp2-6	0	fewer than 2 genes could be modelled
Bbeta	GL541669.1: 56,956–68,458	undetermined	bbr2, bbp2-8	0	fewer than 2 genes could be modelled
Bbeta	GL541733.1: 13,902–30,168	undetermined	bbp2-8, bbp2_b	0	fewer than 2 genes could be modelled

Provenance

GENOME FILE

basidio_microbotryum_
nocall.fna

ASSEMBLY

319 contigs · 25.4 Mb · N50
200.1 kb

GENOME SHA-256

1458786e5fd77f042bed9fdbef6
772b9e590081584e6721228e18b
25fbde9803

MATPREDICT

0.6.1

DATABASE

2b90ce832408b109 · 163
records

TAXONOMY

NCBI E-utilities

PARAMETERS

min_hits=2,
require_core_role=True,
max_polished_clusters_per_family=6,
polish_strong_identity=50.0

RUN STARTED

2026-10-08T04:22:53Z

RUN TIME

197 s

FILES

detection_report.yaml, detected_loci.gff3

Cite MATPredict as given in CITATION.cff at github.com/stajichlab/MATPredict. The reference records named in the gene tables are curated database entries; each record's metadata cites its source publication.

How to read this report

Idiomorph

The form of the mating-type locus a strain carries (for example MAT1-1 or MAT1-2, Plus or Minus). Basidiomycota HD and PR alleles are named after the reference they match.

LIMITS

- MATPredict reads an assembly; it is not a mating assay. Two idiomorphs in one assembly can come from a homothallic species, a mixed culture or a merged diploid assembly.
- Fragmented assemblies can split a locus or lose its flanking genes; an uncalled genome is weak evidence of absence.
- Lineages with no curated record are searched with the whole phylum's references; family labels there are less reliable.